SC452 Cloud Computing Case Study — Source Register: Freshworks Inc.

Course: SC452 (Processes on Networks / Cloud Computing Case Study)
Topic: Freshworks — SaaS Company Built and Scaled on Cloud Infrastructure from India for Global Markets
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Date Accessed / Verified: October 14, 2026

Source Hierarchy Overview

In compliance with research standards, this case study prioritizes:

1. **Primary Regulatory & Financial Filings:** Audited SEC disclosures (Form 10-K, Form S-1).
2. **Cloud Provider Case Studies & Technical Documentation:** AWS official customer success studies and architectural spotlights.
3. **Primary Engineering Publications:** Direct technical posts authored by Freshworks Site Reliability Engineering (SRE), Platform Engineering, and Data Architecture teams.

Organized Source Register

1. Freshworks Inc. Form 10-K Annual Report (Fiscal Year Ended December 31, 2023)

- **Publishing Organization:** United States Securities and Exchange Commission (SEC) / Freshworks Inc. (CIK: 0001544522)
- **Direct URL:** https://www.sec.gov/Archives/edgar/data/1544522/000154452224000012/frsh-20231231.htm
- **Publication Date:** February 16, 2024 | **Date Accessed:** October 14, 2026
- **Classification:** Primary Source (Audited Financial and Regulatory Filing)
- **Specific Claims Supported:**
  - Substantially all production compute, storage, and networking are outsourced to AWS under a long-term commercial contract.
  - FY2023 revenue reached $596.4M (up 20% YoY from $498.0M in FY2022); gross profit was $493.0M (82.7% gross margin vs. 80.8% in FY2022).
  - Cost of revenue includes third-party hosting costs, cloud infrastructure personnel, payment gateway fees, and software licenses.
  - Third-party hosting costs increased by $4.0M in FY2023 (compared to an $8.5M increase in FY2022), with margin expansion explicitly attributed to cloud infrastructure economies of scale and optimization.
  - Note 8 discloses $337.5M in non-cancelable purchase commitments through 2028 ($63.7M in 2024; $67.1M in 2025; $67.7M in 2026; $71.2M in 2027; $67.7M in 2028), primarily dedicated to cloud infrastructure agreements (AWS) and enterprise subscriptions.
- **Relevance & Limitations:** Unimpeachable audited financial baseline for cloud spending and corporate commitments. Limitation: Does not disclose itemized line-item AWS invoices (e.g.,

breakdown between EC2, RDS, and S3).

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2. Freshworks Inc. Form 10-K Annual Report (Fiscal Year Ended December 31, 2024)

- **Publishing Organization:** United States Securities and Exchange Commission (SEC) / Freshworks Inc. (CIK: 0001544522)
- **Direct URL:** https://www.sec.gov/Archives/edgar/data/1544522/000154452225000010/frsh-20241231.htm
- **Publication Date:** February 20, 2025 | **Date Accessed:** October 14, 2026
- **Classification:** Primary Source (Audited Financial and Regulatory Filing)
- **Specific Claims Supported:**
  - FY2024 revenue rose to $720.4M, with gross margin expanding further to 84.3% (gross profit $607.1M).
  - Third-party hosting cost increases decelerated further to +$1.6M for the full year, reflecting enhanced infrastructure efficiency.
  - Future non-cancelable contractual purchase commitments as of Dec 31, 2024, totaled $278.8M ($64.8M in 2025; $72.0M in 2026; $73.6M in 2027; $68.3M in 2028; $0.1M in 2029).
- **Relevance & Limitations:** Verifies that cloud hosting economics consistently improved as customer base scaled past 67,000+ businesses globally. Same itemization limitations as FY2023.

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3. Freshworks Aims for Global Expansion and a $1 Billion Valuation Using AWS Global Passport

- **Publishing Organization:** Amazon Web Services (AWS) Customer Success Stories
- **Direct URL:** https://aws.amazon.com/solutions/case-studies/freshworks-global-passport/
- **Publication Date:** 2024 | **Date Accessed:** October 14, 2026
- **Classification:** Primary / Cloud Provider Case Study (Jointly Authored with Freshworks Leadership)
- **Specific Claims Supported:**
  - Freshworks serves over 70,000 global customers and has crossed $700M in Annual Recurring Revenue (ARR), targeting $1B+ revenue.
  - CTO Murali Swaminathan and SVP Laura Padilla document international expansion into London (UK) and Dubai (UAE) AWS regions to satisfy regional data residency and compliance.
  - Adoption of Amazon Bedrock for deploying generative AI models behind Freddy AI employee and customer experiences.
  - Utilization of AWS ISV Workload Migration Program (WMP) and AWS Rapid Ramp Credit Program for accelerating new feature rollouts while controlling margins.
- **Relevance & Limitations:** Directly confirms geographic infrastructure presence (London, Frankfurt, Dubai, US), compliance motivations, and modern AI architecture. Limitation: High-level executive perspective; does not contain low-level kernel or database configurations.

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4. Scaling the Freshdesk Web App for High Availability

- **Publishing Organization:** Freshworks Engineering Blog / Thoughts on Refreshing Business Software

- **Author:** Bala Kumar (Content Writer, SRE Team Feature)
- **Direct URL:** https://www.freshworks.com/explore-cx/scaling-the-freshdesk-web-app-for-high-availability/
- **Publication Date:** July 15, 2020 | **Date Accessed:** October 14, 2026
- **Classification:** Primary Source (Official Technical Engineering Publication)
- **Specific Claims Supported:**
  - Freshdesk core monolithic Ruby on Rails application serves peak traffic of ~500,000 Requests Per Minute (RPM).
  - Web ingress flow: Amazon Route 53 (subdomain mapping `tenant.freshdesk.com`) → AWS Application Load Balancer (ALB) with TLS termination → HAProxy fleet (1..n) → Envoy Sidecar Proxy → Nginx workers → Passenger Core (single-threaded workers spawned via copy-on-write).
  - Mitigating long-running database transactions: Automated MySQL scheduled event `kill_long_transactions` scans `INFORMATION_SCHEMA.INNODB_TRX` and issues `CALL mysql.rds_kill()` for transactions exceeding 25 seconds to drop TCP connections and trigger rollbacks.
  - Mitigating unresponsive Amazon RDS MySQL shards: Integrated Shopify's open-source Semian circuit breaker gem across MySQL, Redis, and HTTP to fail fast during network partitions and Multi-AZ failovers.
  - Outbound microservice controls: Envoy sidecar proxies manage all outbound calls with connection pooling, retries, circuit breaking, and caching.
- **Relevance & Limitations:** Extremely detailed architectural blueprint of production web ingress, database protection, and resiliency mechanisms. Limitation: Focuses on the core Freshdesk monolith; does not detail standalone newer microservices.

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5. How Freshdesk Uses HAProxy at Scale

- **Publishing Organization:** Freshworks Engineering Publication on Medium
- **Direct URL:** https://medium.com/freshworks-engineering-blog/how-freshdesk-uses-haproxy-at-scale-c6839bc1f784
- **Publication Date:** June 2020 | **Date Accessed:** October 14, 2026
- **Classification:** Primary Source (Official Technical Engineering Publication)
- **Specific Claims Supported:**
  - Implementation of "Shell Architecture" to contain the multi-tenant blast radius across hundreds of AWS web servers.
  - HAProxy sits behind ALBs and routes requests to logical shells based on tenant ID, subscription tier, database shard ID, and URL path (e.g., isolating end-user support portal traffic from agent workspace traffic).
  - Implements dynamic buffer shells: if a database shard degrades, HAProxy isolates affected tenant traffic into a dedicated buffer shell to prevent spillover into healthy customer shells.
  - Uses least-connections load balancing algorithm with round-robin fallback.
- **Relevance & Limitations:** Essential primary evidence explaining how multi-tenancy and noisy neighbors are handled at layer 7. Limitation: Describes routing topology, but specific IP addresses and internal domain maps are redacted for security.

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6. Safeguarding Infra from Runaway Background Jobs

- **Publishing Organization:** Freshworks Engineering Publication on Medium
- **Direct URL:** https://medium.com/freshworks-engineering-blog/freshdesk-safeguard-infra-background-jobs-d56c734c494f
- **Publication Date:** 2020 | **Date Accessed:** October 14, 2026
- **Classification:** Primary Source (Official Technical Engineering Publication)
- **Specific Claims Supported:**
  - Shell architecture bounds database shards and worker capacity, capping each shell at ~60K RPM to isolate disruptions.
  - Integration of GitHub's Freno (replication lag monitor) inside Sidekiq middleware. Before executing database-heavy background tasks, jobs query Freno; if replication lag is detected on MySQL read replicas, jobs are automatically postponed by the lag duration (minimum 60 seconds).
  - Reactive operational controls: Sidekiq middleware allows immediate blackholing (dropping) of non-critical runaway jobs during database CPU spikes or routing to restricted concurrency queues to slash Mean Time to Recovery (MTTR).
- **Relevance & Limitations:** Verifies how asynchronous workloads and database replication are decoupled to preserve SaaS availability. Limitation: Pertains specifically to asynchronous Sidekiq workers.

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7. How Freshworks Engineered a Mission-Critical Self-Hosted Redis Cluster Infrastructure

- **Publishing Organization:** Freshworks Engineering Publication on Medium
- **Direct URL:** https://medium.com/freshworks-engineering-blog/redis-cluster-90cf30934fd0
- **Publication Date:** 2023 | **Date Accessed:** October 14, 2026
- **Classification:** Primary Source (Official Technical Engineering Publication)
- **Specific Claims Supported:**
  - Redis is Freshworks' second-largest datastore (powering Freshchat real-time messaging, Freshservice workflows, Freshsales CRM caching, and background queues).
  - Self-hosted on Amazon Elastic Kubernetes Service (EKS) managed via custom Kubernetes Operators built with the Operator SDK.
  - Architecture features an intelligent Redis Proxy abstraction layer (allowing applications to interface with sharded clusters as a single standalone endpoint).
  - Custom placement and ranking algorithm distributes master nodes and replicas across distinct AWS Availability Zones (AZs) based on historical performance and network topologies, increasing fault tolerance by >50%.
  - Automated self-healing loops for quorum loss and replica failures, accompanied by dynamic Amazon Route 53 DNS record updates during master failovers.
- **Relevance & Limitations:** Demonstrates cloud-native containerized stateful workload management on AWS Kubernetes. Primary evidence of self-hosting vs. AWS managed ElastiCache trade-offs.

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8. The Async Shift: Scaling Our Data Service Without Scaling Our Costs

- **Publishing Organization:** Freshworks Engineering Publication on Medium
- **Authors:** Rahul Misra and Srinivas KNV (Analytics Engineering Team)
- **Direct URL:** https://medium.com/freshworks-engineering-blog/the-async-shift-scaling-our-data-service-without-scaling-our-costs-625e5506770f
- **Publication Date:** 2021 | **Date Accessed:** October 14, 2026
- **Classification:** Primary Source (Official Technical Engineering Publication)
- **Specific Claims Supported:**
  - Freshworks Analytics backend handles over 44 million queries per week (~4,300 queries per minute / ~73 queries per second).
  - Original synchronous query architecture suffered from thread contention, query queue pileups, and escalating data warehouse compute costs as customer load surged.
  - Migrated from a synchronous blocking model to an asynchronous query queuing and decoupled execution engine.
  - Financial outcome: Handled a 5× multiplier in query load without upgrading or expanding underlying data warehouse cluster tiers, effectively decoupling operational volume growth from infrastructure expense.
- **Relevance & Limitations:** Direct primary technical evidence linking cloud cost optimization to architectural redesign. Limitation: Focuses strictly on analytics querying systems.

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9. Freshworks Inc. Form S-1 Registration Statement (Initial Public Offering)

- **Publishing Organization:** United States Securities and Exchange Commission (SEC)
- **Direct URL:** https://www.sec.gov/Archives/edgar/data/1544522/000162828021017717/freshworkss-1.htm
- **Publication Date:** August 27, 2021 | **Date Accessed:** October 14, 2026
- **Classification:** Primary Source (Regulatory IPO Prospectus)
- **Specific Claims Supported:**
  - Establishes founding history (October 2010 in Chennai, India) and global expansion trajectory.
  - Formally defines cloud dependency risk factor: business model is architecturally and operationally tethered to AWS infrastructure, with transitioning to another cloud provider cited as prohibitively costly and disruptive.
- **Relevance & Limitations:** High legal accountability document providing foundational context for Freshworks' cloud-native architectural decisions from day one.

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10. Analyzing Freshdesk Data Using Amazon EventBridge and Amazon Athena

- **Publishing Organization:** AWS Compute Blog
- **Author:** AWS Solutions Architecture Team
- **Direct URL:** https://aws.amazon.com/blogs/compute/analyzing-freshdesk-data-using-amazon-eventbridge-and-amazon-athena/
- **Publication Date:** August 2020 | **Date Accessed:** October 14, 2026
- **Classification:** Cloud Provider Technical Documentation
- **Specific Claims Supported:**

    - Details event-driven integration between Freshworks SaaS and native AWS serverless services (Amazon EventBridge, Amazon Kinesis Data Firehose, Amazon S3, Amazon Athena) for streaming audit trails and ticket lifecycle telemetry without maintaining polling infrastructure.
- **Relevance & Limitations:** Demonstrates deep API-level hooks between Freshworks and AWS ecosystem services.